

Search sheets: how to run the activity



The pages after this one are 36 **search sheets**, built from *The Cat in the Hat* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

If the room is a different size, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 54–55 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **in** **the**, the boxed token is the previous token and **the** is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 8 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *the*? it’s a purple one” — and most of the page stops being worth reading. The swatches are chosen to answer to these words, so the words are the ones to use; the sheets themselves print no key, because a strip of swatches at the top of a page of swatches is one more thing to scan past.

black **grey** **red** **brown** **green** **blue** **purple** **magenta**

Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Quiet rounds. One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

1962 tokens, 240 unique · 2.83 bits/token · perplexity 7.1 · branching factor 4.32

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token *and* its next token on the board.

Round 1. Call out the last token on the board (**the**). Another sheet has a match, so that person raises a hand and reads out *milk*.

Round 2. Call out the last token on the board (**milk**). Another sheet has a match, so that person raises a hand and reads out *on*.

Keep going for as long as you like. The text the room writes will sound like *The Cat in the Hat* without ever repeating it — that is the model generating, not the corpus being read out.

the token pair that answers

in the

the milk

milk on

on the board

in the

in the milk

in the milk on

in the	game that	fish in	said ,
things will	Why do	what would	. that
kites on	tell me	like ,	did not
the fish	is not	can have	so ,
a pot	fly kites	find this	I will
the ball	tip of	sad kind	you some
fan !	us this	, wet	! so
have to	do as	should not	! look
wet to	had to	saw them	with the
and I	bump on	. have	I do
he said	we do	that is	milk on
the way	. and	then I	and so
to you	! then	not mind	, I
will show	on a	. not	

find us	did you	Why ,	not all
! and	and Thing	day .	and the
so tame	, white	said the	hands with
a cake	, what	! said	is a
shame !	wet day	kites ,	I call
. .	. your	with my	then we
can hold	put them	I do	. your
we had	have Lots	. he	tip of
to us	the milk	will not	saw him
you do	the cup	mess ,	on the
fish ,	like to	he put	fan as
do something		house and	! said
that is	the fish	milk and	him !
what to	call up	about it	things and

then , him step mat ! fan ,
 put me . you what will house .
 ? Oh you can call them on a
 I said deep and wet , the cat
 ! and can hold to us . we
 and Sally have any ! Thump hop up
 with a things I ! Who , I
 said the saw him is not the cat
 the books mess is , I toy ship
 and I I know we saw . and
 he said like it hands . milk ,
 not a will see . he about with
 that are day . shame ! fish !
 so so bad tricks do !

what to	him pick	! said	is not
the hat	. Yes	fish .	. two
house !	call this	with a	, with
are pink	said the	want to	toy man
so all	put them	shook their	hands with
you ?	will like	! look	fan with
bad ,	day .	wet and	like these
can hold	? Well	shame !	mess !
I will	! so	on there	he fell
we did	and so	do ?	about .
not know	that is	then Sally	. that
the hat	saw one	to find	and big
things fall	. do	hop on	, said
I will	the cake	deep in	

! what	things they	our fish	get them
gown with	fish in	then Sally	. you
wet day	house !	lit .	! and
the toy	toy ship	not sunny	to us
want to	day ?	I can	some good
we looked	he gave	on his	that bump
would you	what a	and Sally	. so
put me	so .	do not	I say
! now	good fun	will show	saw him
the cat	Think of	, said	a ball
can hold	said to	here .	like it
with a	the house	shook up	, no
you should	? Why	and we	are two
. .	there like	. Thing	

. then our fish ! and good ,
 on a house ? and down . I
 Sally . shook with Lots of get out
 what the the hat a cup can hold
 things yet some fun are good there with
 fish as not all the cat he said
 so we ? tell toy ship ! I
 . But will not . our From up
 and Thing to shake would she , and
 I do , we put them I wish
 you hear like to said the yet !
 then I gown ! that . we saw
 the hall kind of now you ! bump
 here When not be at me

Sally and pick up that is ! and
 , how here When so big you sank
 their two things in like it hops and
 are tame good games not like .
 what did , and said the get those
 then those fish said ! sit they will
 house ! our toy not all Lots of
 to be the cat we saw .
 ? and a cup he came some new
 . I on the and Thing ! he
 . and I like there , would you
 the cat at all gown , now look
 bit ! of her can hold I said
 and this will she the hat

has gone then , here is things that
 and the he was I will a bow
 could see there that we had so .
 will hit be in at this not like
 I will . what . my ! then
 can get that is ! we they will
 their kites bit ! on the our mother
 said the not be ! you some milk
 are not fish said pick up and the
 ? I like that good tricks what to
 now ! you wish . I , wet
 the fish of the get rid the cat
 house ! to get , too bite you
 . we Sally and the cake

tail I	that is	good trick	he lit
gave them	things .	like to	dish !
of this	be here	cold to	can fan
! then	and the	PLOP !	now !
what you	on my	not .	, not
here to	they play	we saw	. .
? what	at all	! said	then our
not shine	house !	a pat	hear !
our mother	to fall	there on	you up
could ,	the rake	the cat	said the
fish .	pick it	I can	I will
. he	so we	. But	, said
. and	Sally and	will have	For the
bit .	! those	the fan	

, out	he got	to do	into a
hear ?	! the	a look	, then
! and	and I	our fish	fish shook
! what	were down	pick up	dish ,
things .	I will	not in	could do
Sally and	there is	the hat	be here
here .	at me	on the	the fish
they like	? should	in the	will show
how do	fast as	now !	house all
. she	that were	of this	. .
cold ,	I do	not fly	the hook
said ,	pack up	can get	! Oh
like this	what will	good game	. .
we can	then he	you pack	

we saw a rake fast as of good
 in the can hop will not . he
 be here kicks and not be books !
 up and ! I on the man !
 and all , she things ran I can
 could play our fish kinds of give you
 , what that I cold , . we
 fish . you sit ! I cat go
 . and string of house For how that
 what would I hop the cat the dots
 ? Oh good things said to to know
 then something now , here When
 he shut . . at that Sally and
 like the the house ! look they like

said to	can have	. these	be here
, we	things they	? now	one and
in with	two things	the day	what a
and the	not want	how I	on the
at me	will give	strings ,	to the
a book	then he	! so	and Thing
! if	box ,	I ,	now look
playthings and		books ,	like it
up these	of something		the pot
mother's new		! they	. then
. and	our house	fish in	. I
we saw	net .	wood box	fast .
, so	you like	that I	lot of
I can	show you	cat in	the ball
they are	good fun		

I call	show to	I can	box are
like our	one little	the house	our new
a tip	, said	! said	to say
you another	trick that	little toy	, the
! her	now ,	net .	in the
! put	out of	, Why	. this
cat in	lot !	fish said	what a
of good	. it	. and	they should
pat .	two books	. Oh	good game
things that	said the	not be	we sat
how .	will not	the box	play .
books !	at all	be here	and kicks
then he	fast as	that day	and so
? would	up the	the fish	

, with	. so	. I	to the
the things	then he	trick ,	fast !
out and	we can	net ,	last !
will come	looked !	they should	the hat
Thing one	and a	Oh ,	two things
! do	I can	fish came	pot !
tricks .	one and	the fish	cat in
said the	. a	in the	, do
a box	our mother	and look	now .
at last	! and	be about	box .
I do	you have	! said	that is
up on	, said	show them	not one
came down	play !	no fear	little fish
. this	? Oh	of look	

no ,	. I	I said	show you
up those	box with	you do	we sat
. have	pot .	will bump	the cake
then our	the mat	two ,	Oh ,
that they	out !	little bit	not wish
Thing two	. .	the ship	and red
looked !	now look	one one	! at
, no	said the	of his	in the
, and	and I	net I	funny !
sat in	, here	they will	cat .
came in	our fish	at me	! those
hall .	! put	sun did	a bump
fish .	play ,	tricks .	to go
. they	I can	? and	

of Thing	Oh dear	and the	. there
our fish	box .	in the	. I
little bit	! how	! then	two .
them go	! then	not .	one and
. with	cat in	said ,	play ball
pot .	, what	tricks ,	no fear
Thing two	bow .	and you	hall .
out !	take them	will find	the cat
we sat	this mess	now !	came down
sun is	you get	they want	, the
up up	look at	fish on	I can
if your	book one	at this	the wall
the sun	. that	I let	a lot
, I	to do	sat there	

, jumps not pick . you our game
 look ! if you to do and the
 in a and a take a fall .
 When your of one , how do not
 we looked they ran play . ! said
 the hook the cat little bit two things
 fear . . I the milk at all
 this mess said the came down , look
 a big them a no fun I do
 out of ! have home ! I saw
 box came ! I Oh , one hand
 up high pot . run down sat there
 Thing one his hat cat . tricks are
 you will fish to . it

up .	at me	! they	cat .
When our	in the	the cat	came Thing
, now	of the	hand !	the cat
and Thing	. .	. and	with Thing
to the	, I	sat in	no fear
you now	we tell	Oh ,	the wall
those things	I call	new tricks	play .
do ?	they have	look at	asked you
one !	his hat	I always	said the
fear !	them out	, said	. no
out !	fall !	Thing one	if I
not know	! then	little toy	a little
pot .	! said	bed !	two things
and I	this game		

Thing one new . at this a tip
 came back this way Well . we saw
 the house , and out of not be
 . they When your said , cat .
 went on and Thing his head , said
 and they I had fall ! look at
 no fear ! I fear ! ! now
 . I . in is what ! he
 Make that one ! up all the cat
 with a you do of my do .
 Oh , tame . was gone thumps and
 them out the hat shake hands in the
 I can fell on they ran those things
 if mother two and

cat with	said the	out .	a fish
! put	! .	said our	tame !
those things	! and	I bet	went away
cake !	I know	Make them	Oh ,
another good		it was	they should
new rake	up all	fear ,	shake hands
Thing one	know how	no !	, that
the fish	and the	fell into	. I
wall !	with big	not know	do not
. you	look at	this house	and you
the things	! and	of his	in the
, said	them I	one little	two !
is not	the fish	was out	mind at
you did	say .	we could	

! they , said sank it Oh ,
 my tricks cat , Thing two .
 the things was to say . and our
 they said two ? the hat . and
 the pot , and up the a box
 with hops is not fear of sit !
 said the of good I stand went bump
 look ! bet . ! have ! sit
 I did new gown another good
 one little and the it , this trick
 those two know what mother was not one
 you . all the down on do not
 them Thing wall in in the cake .
 said the . I no fear

one kite no ! do . as I
 Oh , they ran all . me !
 I sat my tail . they look what
 them to the cat sank our not let
 . should fear . said the ! bump
 the cat come here , as and he
 is not , sank ! it went after
 was too her the two things and so
 up there wish we with a say .
 sit ! . your cake , know !
 I like cat . of bad those two
 it deep bet , in the Thing two
 said , this wet ! not down the
 mother could a red

was shut . so then come .
 those things ! what sit ! say ?
 step in ! and is good up our
 cat went wish to and some I know
 the string down with me . them away
 bumps , one kite I went , no
 a dish two . do you as I
 cake on mother is look ! with a
 to have , and ! your the hat
 big red not ! hat . this box
 all kinds fun to and our . now
 it a my playthings no !
 her . said , Thing two know it
 Oh , way that in this

the fish the cat to play cat !
 and look . and not bad and he
 said , in the away . it was
 a sad all we ! and them away
 sit there way ! high as my hat
 her way mother is , if , said
 ! down Oh , you will do I
 is near fun in no way as fast
 I could was back Who was gone with
 down my up up say ! one and
 me now ! said . you hat !
 two . this is big and with a
 Thing one know some . I wish ,
 pink , look at

picked up , we to us , wet
 mother ! is wet . look as a
 you take gone away no ! up !
 ! look her about look at this !
 them run two and the house and the
 fun on a house I did . Oh
 down ! hat you Oh , away !
 cup on know ! ! said ! Make
 the cat ship , Thing two me !
 said . my net But that these things
 wish to way home it came all !
 and he . with nothing at in and
 cat . sit ! not bite do you
 big bumps with a

mother will ! fast a game all .
 , and up with tall , stand on
 fun at ! said hat . know what
 do not in the way at them !
 big thumps said the not like no .
 two hands cat in and the the hall
 I will dear . . Oh . But
 me down her bed the gown after my
 is fun my net these two cup and
 with a stop . . a it is
 Oh dear this ? you hear , no
 But you and the away . look !
 Yes . down the to play ! But
 ship ! as I

. look it ! I like down !
 do was is that dear ! fun ?
 ! no away ! and then ! have
 that cat something went my net
 a good , said . . hat .
 ball ! her gown ship , said the
 But your look ! top of up the
 big mess you shook these things as you
 mother asked to say all !
 have to . . not one the top
 cat . me down know some ! we
 let you should we with the cup ,
 the cat and I , did back in
 in a rake . this is no !

this house should not a game fun But
 it is , said have no ship and
 my net hat . ! and me !
 no . down . let down cat in
 ! said the fish she will ! I
 know what look at all ! and I
 . and away with ! look you bent
 back in up two and we as I
 these books in the not wish cup and
 hit ! I did But I . then
 said the is not mother will do you
 the cat top with , we rake ,
 with a ball . to stop . I
 that they something to

in the ! and all . ! then
 ball . ran up to play a shame
 my hat and the hat to , as
 ! and is so did not rake !
 kite bump this , you . she do
 hook . look at that is down ,
 . we up the fun that and Sally
 your mother me now should not sunny .
 cat . with the as he no !
 said the the things have no I can
 . . it to But that do ?
 something new ! too go !
 . they not all the sun any fun
 mother came away . , with

so , the cat go away . now
something to should not do !
. take and she , Oh hold up
the box in on she say hat ,
kite we cat in ball . up the
Thump ! have fun But our hook .
! I ! sit with my I saw
got up it . mother is bump !
ran down a PLOP said the is on
did nothing to fly down with .
! look ! I fun , look .
not . your mother , and that I
you do this house as fast he should
too . all the and a

. But	, little	have no	! and
bump made	. it	and I	is out
fun .	in the	to you	do not
all .	he picked	! on	with a
should not	that !	so deep	did you
I know	too wet	it ?	we can
. they	she will	cat .	! then
! put	mother is	your mother	games we
you do	not like	and a	hook .
look at	a little	something !	ran to
Thump !	the things	always pick	head !
, and	go away	said the	ball !
down !	But that	this big	hat !
the cat	hold you		

and then	, cold	not like	! no
things had	said the	it up	you fall
will hold	all I	jump !	cat in
ran about	he should	game that	the pot
on top	near !	to sit	with Sally
! Oh	hold up	should we	she say
! you	bump !	your mother	. Oh
did not	go out	that is	mother is
and with	. he	. he	hat .
we shook	shut with	. now	so so
too cold	head of	in the	, and
then we	! and	have Lots	do to
us two	I said	But we	a shame
is no	the ball		

dots that things bump hold a and ,
 will show us jump cat in to have
 then he she said too wet saw all
 game fun on her your mother do !
 said the the cat ! they have no
 a fox ! tell bump From you have
 should we with my , fast . you
 we two red . hat . he should
 , no is no . you . they
 I bet . But go ! ! But
 and the the fish in the jumps and
 that ? shut up But our did not
 not like the things so we ran out
 all that ! with

will pick should not your mother us ?
 tell her do you I do to shake
 bump ! on the with fear you another
 then ! the fan in this that cold
 game . red wood ! Make , I
 fox , fish in and too cat .
 have fun not be had something
 a hook hold up said the , Think
 rid of is not we did ! Oh
 made us . that ! said see this
 . I he fell so tall ! Thump
 saw those the fish things should
 can not hat ! . then all if
 did ! the cat and then . Why

not go	white and	the hat	that cat
your mother	! and	cat in	I know
fish .	and the	then he	will she
so do	all my	he ran	! they
did not	on this	said ,	to play
tell her	, the	the strings	bent our
red fan	should not	! I	can ,
us fast	. then	. Oh	and I
hold up	the pot	wet to	. then
a shame	hat came	game ,	bump their
you did	had them	do if	have no
. I	saw mother's		with a
things are	see something		we could
is funny	in the	! Oh	, said

so , have come on the . these
 , so the head and I I can
 . what do not and then hold the
 cat said kites in he has ! But
 bump ! we tell should not us ,
 not all Why , what should saw her
 tell that things , ! not to go
 fly kites ! now game that fish !
 , the did not that went can do
 . no then we said , shine .
 with Thing the dish like it in the
 will not a lot the cup . so
 you will is not your mother had to
 hat . wet ,